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Strengthening nucleic acid biosecurity screening against generative protein design tools

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Advances in artificial intelligence (AI)–assisted protein engineering are enabling breakthroughs in the life sciences but also introduce new biosecurity challenges. Synthesis of nucleic acids is a choke point in AI-assisted protein engineering pipelines. Thus, an important focus for efforts to enhance biosecurity given AI-enabled capabilities is bolstering methods used by nucleic acid synthesis providers to screen orders. We evaluated the ability of open-source AI-powered protein design software to create variants of proteins of concern that could evade detection by the biosecurity screening tools used by nucleic acid synthesis providers, identifying a vulnerability where AI-redesigned sequences could not be detected reliably by current tools. In response, we developed and deployed patches, greatly improving detection rates of synthetic homologs more likely to retain wild type–like function.

A growing array of generative AI tools is being harnessed to design new proteins or modify existing ones, enabling scientists to cut through combinatorial design spaces to achieve specific structural and functional goals (*1*–*7*). These AI-assisted protein design (AIPD) tools hold significant promise for addressing some of humanity's most pressing challenges in the health and life sciences (*8*). However, given their rapid pace of advancement and increasing accessibility, protein scientists, biosecurity experts, and regulators are recognizing the need to develop protocols and systems to guard against their misuse in producing potentially harmful proteins (*8*–*26*).

A critical step in the synthesis of any harmful protein is the ordering of a gene (or fragment thereof) encoding that protein from one of numerous synthetic nucleic acid providers. And, although obtaining such a gene would only be the first step among many toward developing harmful proteins (whether inadvertently or with malevolent intent) (*15*, *16*, *23*), it is an all but necessary one, creating a bottleneck in the protein production pipeline that can be monitored to reduce the risk of such proteins being obtained by inappropriate actors. Indeed, nucleic acid synthesis companies use biosecurity screening software (BSS) to screen customer orders for genes that encode proteins of concern (POCs).

Most BSS systems in use today operate by searching a sequence for fragments of controlled nucleic acid or protein sequences. If a fragment is a better match to a sequence of concern than to any other, then the sequence is flagged and queued for expert review. This process, referred to as "best match," was originally defined in the since-updated sequence screening guidelines published by the US Department of Health and Human Services in 2010 (*27*).

Modern BSS systems are the result of refinement over time to improve resilience to newly emerging exploits. For example, it was once challenging to change the wild-type nucleic acid sequence encoding a protein while maintaining protein expression, but the emergence of codon optimization and reassignment tools allowed for the design of nucleic acid sequences with low identity to controlled ones but with otherwise identical amino acid sequences. In response, BSS tools were made codon resilient by adding six-frame translation of nucleic acid sequences and checking for sequence similarity to amino acid sequences of concern (*27*–*29*).

From the range of possible interventions suggested for safeguarding against AIPD-enabled threats (*12*, *13*, *15*–*18*, *21*, *22*, *30*), continued and improved investment in BSS is among the most frequently suggested (*26*, *31*–*35*). However, as we show in this work, best match–based BSS is itself vulnerable to AIPD-enabled exploits: In a manner similar to the codon exploit, AI methods using protein sequence generative models (PSGMs) enable bad actors or unsuspecting researchers to change the amino acid sequence of a protein while pursuing maintenance of function. In doing so, the resultant "synthetic homologs" can have limited identity to a controlled amino acid sequence and thus can be difficult to detect (*2*, *36*–*38*).

Despite concern among experts that traditional BSS may be less sensitive in the face of such AIPD capabilities (*10*, *11*, *39*), no large-scale evaluation of potential vulnerabilities has been performed to date. In this work, we report a multimonth, cross-organizational effort to pursue insights about this challenge and to develop mitigations. Our study centers on an "AI red teaming" analysis (Fig. 1), showing how freely available AIPD tools can be used to perform adversarial AIPD with the goal of evading BSS. We also show how resulting findings can be used to guide mitigations.

Framing study and disclosure process

Our project was stimulated by an initial end-to-end adversarial AIPD framing study performed in October 2023 in preparation for a forthcoming summit of protein engineering scientists on biosecurity at the University of Washington (*40*). In the framing study, widely available AIPD tools were used to demonstrate in silico how a POC could be reformulated into thousands of synthetic homologs, and how those sequences could not be detected by the BSS used by two prominent nucleic acid synthesis companies (appendix S1).

As discussed in greater detail in the supplementary materials (see considerations and process), we decided to treat the discovery of this BSS vulnerability as a type of biological "zero day," a term commonly used in the cybersecurity domain to describe software or hardware vulnerabilities that were previously undescribed and for which no patch exists at the time of discovery (*41*). Taking inspiration from established cybersecurity processes for addressing such situations, we contacted the relevant bodies regarding the potential vulnerability, including the International Gene Synthesis Consortium (IGSC) (*42*) and trusted colleagues in the protein design community as well as leads in biosecurity at the US Office of Science and Technology Policy (OSTP), US National Institute of Standards and Technologies (NIST), US Department of Homeland Security (DHS), and US Office of Pandemic Preparedness and Response (OPPR). Outside of those bodies, details were kept confidential until a more comprehensive study could be performed in pursuit of potential mitigations and for "patches" to BSS to be developed and deployed. The results of that expanded study are presented in the remainder of this manuscript. Additionally, now that the vulnerability has been addressed, we have made the framing paper from October 2023, which was initially only shared with the above-mentioned groups, available as an appendix to this manuscript (appendix S1).

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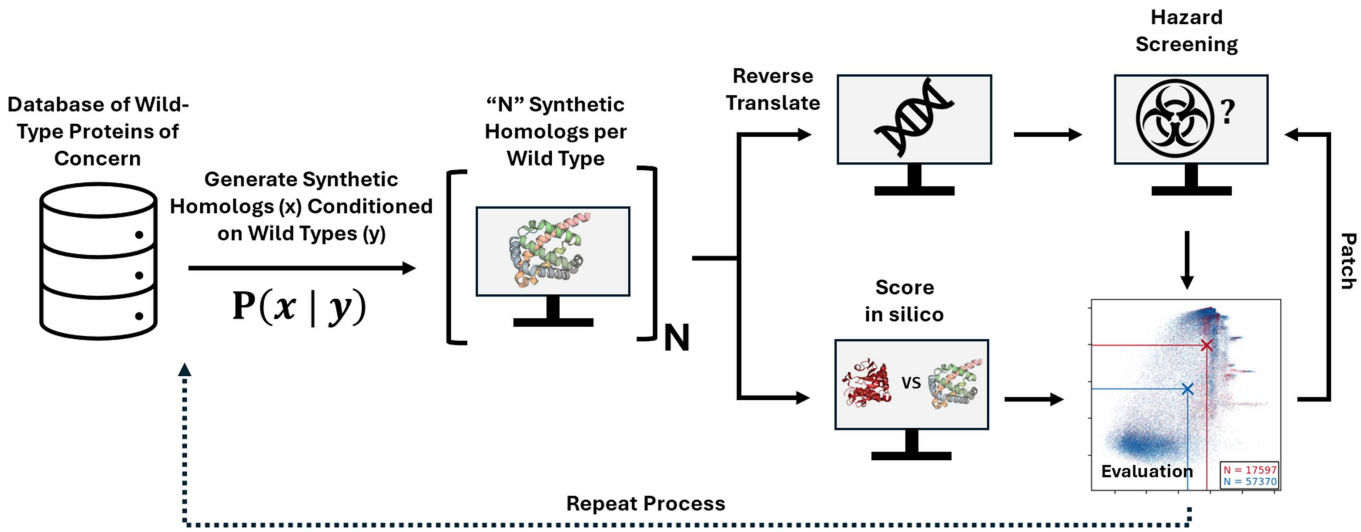

Fig. 1. Summary of AIPD red-teaming workflow. (See also table S1.)

In silico protein reformulation pipeline

We began our follow-on investigation by using three open-source PSGMs [ProteinMPNN, EvoDiff-MSA, and EvoDiff-Seq (*6*, *43*)] to generate 76,080 total synthetic homologs across a spectrum of mutational loads for 72 different wild-type POCs (table S1 and supplementary materials, reformulation). The 72 template proteins are primarily toxins, as obtaining a single toxin sequence can be sufficient to constitute a threat, whereas, individually, nontoxic genes from controlled organisms would require additional context (e.g., a complete genome) to produce harm, so they can be considered a less salient risk. That said, a small number of proteins from controlled viral species were also included to ensure that results would not be overly specific to toxins. Note that outside of table S1, we refer to each of the 72 template proteins by a randomly assigned index, as we believe that directly linking protein identities to results would constitute an information hazard.

We did not construct or test any of the proteins generated as part of this study in a wet laboratory setting, having been cautioned to remain cognizant of potential negative perceptions (international and otherwise) of such work being interpreted as pursuing the development of bioweapons. Such confirmatory work and real-world calibration through lab-based studies, if conducted, must be carried out with appropriate security, transparency, and oversight. Instead of wet-laboratory testing, we used OpenFold predictions of template proteins and synthetic homologs to calculate the in silico metrics "TM-Score" and "ΔpLDDT" (Fig. 2 and figs. S1 to S6; supplementary materials, evaluation) (*44*). Literature evidence suggests that in silico metrics can function as a noisy proxy for the likelihood of a synthetic homolog maintaining function, and, consequently, that at least a subset of the 76,080 synthetic homologs would retain function (supplementary materials, literature review: in vitro success rate of generated proteins) (*1*, *2*, *4*, *5*, *7*, *36*–*38*, *45*). We stress, however, that the goal of our red-teaming effort was to improve the ability of BSS to detect malevolent intention through identifying potential sequences of concern, not necessarily to detect confirmed functional sequences of concern. With this goal and context, we believe that in silico metrics are sufficient for evaluating the relative performance of BSS implementations.

Evaluating biosecurity screening software

The 76,080 synthetic homologs were sent to four BSS developers, who subsequently ran the sequences through their software. The BSS tools tested span a range of strategies, from regular-expression

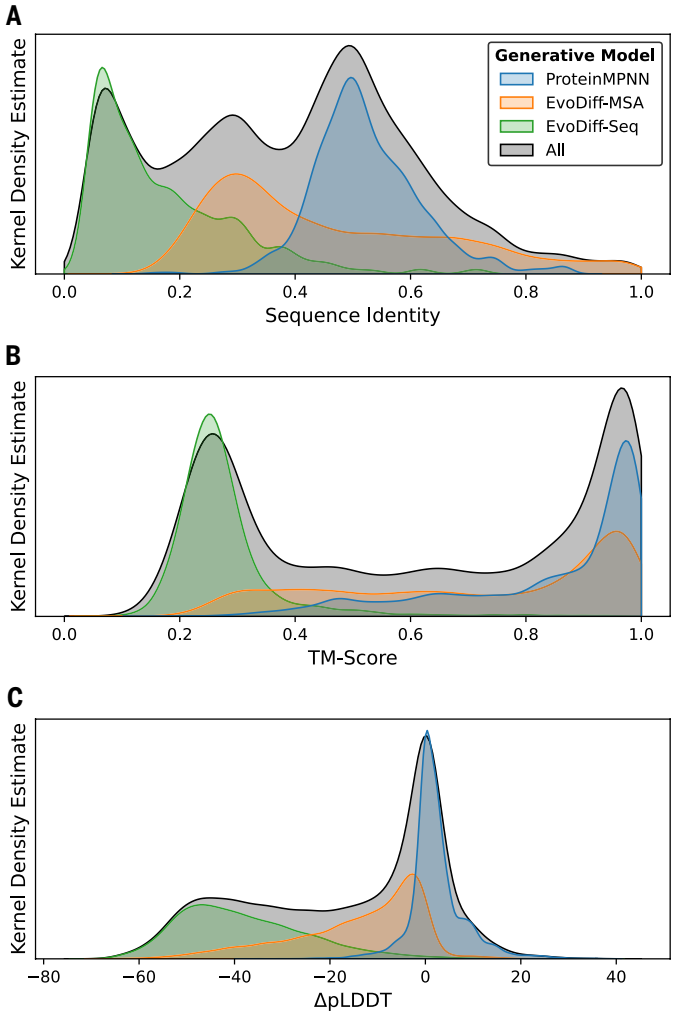

Fig. 2. Distribution of metrics for the synthetic homologs generated in silico by ProteinMPNN, EvoDiff-MSA, and EvoDiff-Seq. (**A**) Distribution of sequence identities between the wild-type proteins and the derived reformulated variants. (**B**) Distribution of TM-Scores between the wild-type proteins and derived reformulated variants. (**C**) Distribution of the difference in pLDDT between the wild-type proteins and the derived reformulated variants. See also figs. S1 to S6 and data S1.

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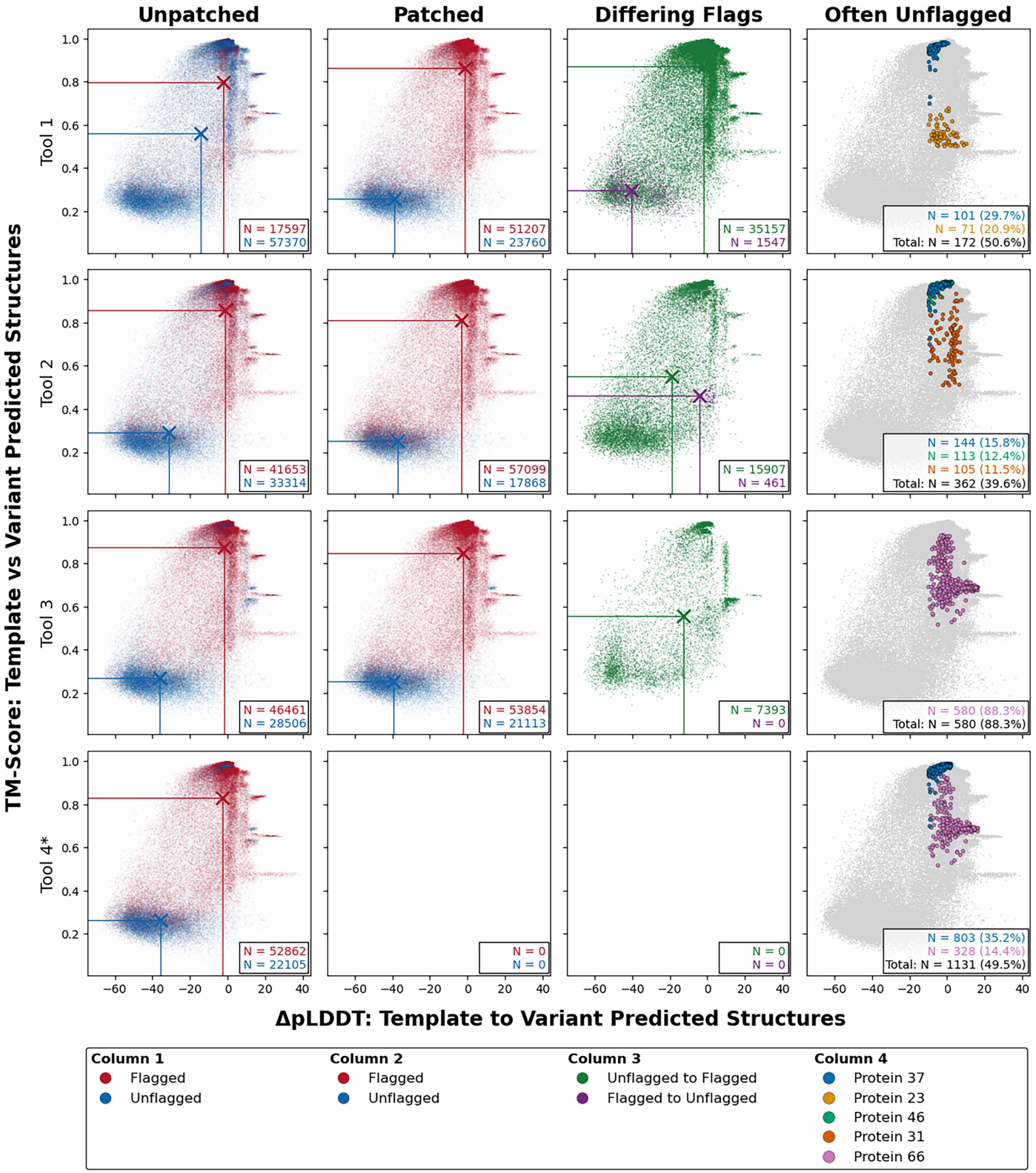

Fig. 3. In silico metrics and flags for all proteins, tools, and participating providers. Columns 1 and 2 give results for the unpatched and patched versions of each tool, respectively. Column 3 gives the changes in flags between Columns 1 and 2, with green dots for proteins that were not flagged in Column 1 but that are in Column 2 and purple dots for the opposite. Proteins whose flags did not change are not plotted in this column. Grouped by template protein, Column 4 highlights the synthetic homologs that have high in silico metrics (TM-Score > 0.5 and ΔpLDDT > −10) and were the most frequently unflagged by patched tools. Boxes in the lower-right corner of plots in this column give the number of these proteins and the percentage of all unflagged high–in silico metric "misses" that this number constitutes. Only proteins constituting >10% of misses were highlighted in this column. See also figs. S7 to S18, tables S1 to S3, and data S1 to S5. *Note that no updates were made to Tool 4 in this work, hence the corresponding empty plots for Columns 2 and 3.

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matching to Hidden Markov Model–based search and neural network–enhanced sequence alignment (supplementary materials, biosecurity screening software), though all use the best match framework and thus might be expected to struggle to identify the adversarial sequences generated here.

The results of runs performed by using BSS with screening protocols in use at the time that the framing study was performed can be seen in the first column of Fig. 3. A synthetic homolog with a high TM-Score and high ΔpLDDT is expected to be more likely to maintain the function of its wild-type template (supplementary materials, evaluation), so, in this column and the second, it is important for BSS to flag the majority of the sequences in the top right of the subplots (i.e., the top right of the subplots should be red). It is also important to note that, when applied to standard nucleic acid synthesis orders, most of which will not contain potentially hazardous sequences, the false positive rate must be kept low to avoid unnecessary delays in order fulfillment and minimize the economic burden imposed by screening on the nucleic acid synthesis industry. Thus, we do not necessarily want to see BSS flagging proteins in the lower-left corner of these plots. Still, given the potential for the proteins studied in this work to cause harm as well as the uncertainty inherent to our in-silico metrics, it is more important here to see the upper right colored mostly red (flagged) than the lower left colored mostly blue (unflagged). To be most confident in a BSS system, it should not include blue patches in the upper right of the first or second columns.

Overall, the results shown in Fig. 3, Column 1, were mixed, with some high–in silico metric proteins not flagged by all tools, but the overall number of misses varying by BSS. This was not the case when these protocols were applied to the wild-type proteins used to build the reformulated variants, however, where we observed near-perfect sensitivity and specificity for all tools (tables S2 and S3). Together, these results indicate that, although all tested BSS tools worked well for wild-type proteins, given the newfound capabilities of PSGMs, the BSS tools were not consistently capable of identifying DNA synthesis orders that contained genes encoding potentially harmful proteins.

Capabilities of updated screening tools

In response to the results reported in the framing study and/or the results presented in the previous section, Providers 1 to 3 made updates to their BSS, ranging from changing the screening algorithm entirely (Provider 1) to simply updating parameters to improve sensitivity (Providers 2 and 3) (supplementary materials, biosecurity screening software). Provider 4 did not perform updates to their tool for reasons discussed below. As can be seen in Column 2 of Fig. 3, after making changes to their BSS strategies, Providers 1 to 3 became more capable at identifying AI-reformulated variants and, in some cases, also at identifying the wild-type templates, all while keeping false positive rates low (tables S2 and S3).

Despite showing bolstered screening capabilities, none of the patched BSS tools could identify all proteins with high in silico metrics. If we define a "more-probably-functional" protein as one with a TM-Score above 0.5 [a value typically cited as a threshold for whether two proteins are likely to share the same overall fold (*46*)] and a ΔpLDDT >−10 (a more arbitrarily chosen cutoff that, when combined with the TM-Score cutoff of 0.5, isolates the cluster of proteins in the top right of Fig. 3, columns 1 and 2), then Tool 1-patched does not flag 340 (~1%) of the proteins, Tool 2-patched does not flag 913 (~3%), Tool 3-patched does not flag 917 (~3%), and tool 4 does not flag 2287 (~7%), corresponding to an average of ~3% of sequences not being flagged over all tools (or, inversely, a 97% flag rate).

We again stress that we should not expect all (nor even, necessarily, a majority) of the more-probably-functional proteins to actually be functional. The cutoffs were selected to simulate the behavior of an adversarial actor, who would, as is common practice when selecting generated protein sequences for analysis in vitro, likely use an in silico prescreen to filter candidates. Indeed, follow-up work that we performed using benign proxies of proteins of concern while this manuscript was under review suggests that proteins with more complex functions are unlikely to function at all (*47*), a somewhat comforting result for the short term, yet one that is unlikely to hold as PSGMs continue to improve.

Despite the uncertainty regarding functionality, it is still important to analyze why some more-probably-functional sequences were not flagged (herein referred to as "missed sequences"). We found that most of the missed variants are derived from only a few template proteins (figs. S10 to S18): Proteins 23, 31, 37, 46, and 66. Synthetic homologs from Proteins 31, 37, and 46 are derived from POCs with high sequence similarity to proteins that are not of concern. If a reformulated variant was a better match to one of these nonconcerning proteins than to a concerning one, then it was not flagged, which is considered the correct outcome for a BSS tool under existing guidance from the US government. For Protein 66, the coauthors disagreed about whether its synthetic homologs should be flagged at all, as Protein 66 (i) is only an accessory component of a controlled toxin that provides basic structural integrity with some potential (though nonessential) role in toxin entry and (ii) comes from a toxin type that is not typically attributed to human disease. Lastly, variants of Protein 23 (frequently missed by Tool 1) were not detected, as the uniform parameters used for training the patched version of Tool 1 in this study resulted in poor detection of these proteins. Subsequent adjustment of the parameters for this template addressed the problem (results not shown).

The causes of misses highlight some core learnings from this study: (i) The high miss rate for Proteins 31, 37, 46, and 66 highlights the necessity of agreeing on what constitutes a potentially hazardous sequence, especially considering that AIPD can populate the grey areas between clear positives and negatives (*48*, *49*). Indeed, Provider 4 opted not to update their software precisely because it was unclear whether their tool's "misses" should have been flagged in the first place. (ii) The lack of flagging of variants of Protein 23 by Tool 1 shows that, although an approach may generally work well, rather than a one-size-fits-all approach, at least some degree of customization may be needed for detection of specific classes of threat.

Challenge of obfuscated AI-reformulated proteins

As a second set of experiments, we presented a patched BSS with obfuscated DNA created by (i) fragmenting DNA sequences directly encoding synthetic homologs and then (ii) shuffling the resulting fragments' sense and/or order (supplementary materials, reverse translation). Such a strategy is a well-known exploit that can be used to hide a protein's identity from BSS while still enabling a bad actor to obtain a protein-coding gene. Shuffling makes it impossible to directly translate the nucleic acid into the protein it encodes, thereby making detection more difficult; however, in a laboratory setting, standard molecular biology techniques could be used to rebuild the original gene (*39*). We included this test to evaluate the ability of updated BSS to detect genes encoding POCs created by a determined actor employing multiple adversarial techniques in combination.

Figure 4 shows the frequency with which more-probably-functional reformulated variants from each of the 72 template proteins were obscured by this obfuscation process, where an "obscured protein" is one that was flagged when directly encoded by DNA but not when encoded by obfuscated DNA. Overall, relative to unpatched BSS, we see that, along with having a greater ability to detect AI-reformulated variants, all patched BSS tools have an improved, though not absolute (table S4), resilience to DNA obfuscation as well. BSS detects fragmented POCs by screening for subsequences of concern, so it is not surprising that some obfuscated variants are still missed by the patched tools, as both reformulation and DNA obfuscation reduce the number and length of contiguous regions of sequence mapping

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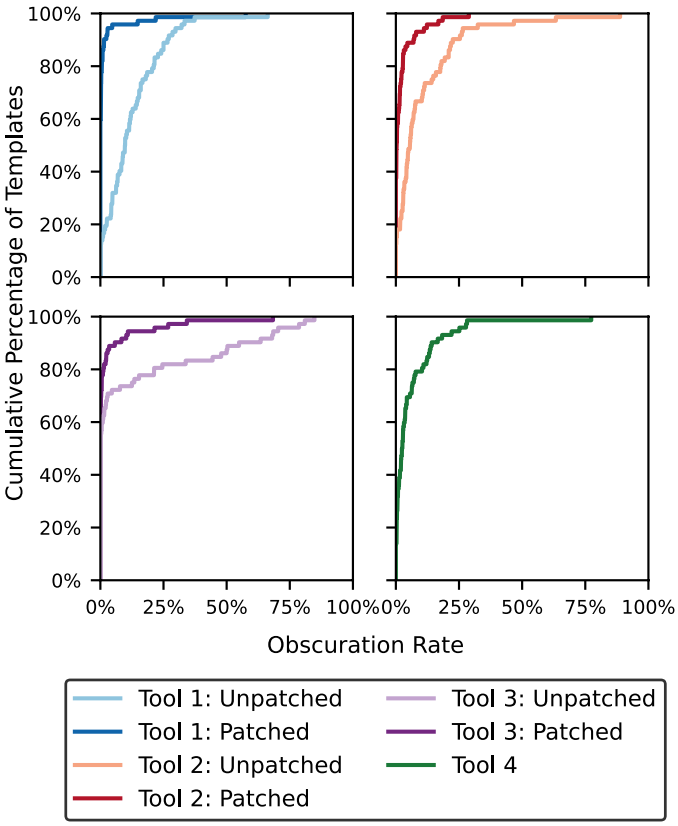

Fig. 4. Effects of DNA obfuscation on BSS for synthetic homologs with high in silico metrics (TM-Score > 0.5 and ΔpLDDT > −10). Each subplot gives results for a different BSS provider as empirical cumulative distribution functions of the frequencies with which reformulated proteins were obscured by DNA obfuscation, where an "obscured" protein is one that was flagged when directly encoded by DNA but not when encoded by obfuscated DNA. The *y* axis indicates the percentage of templates that have a rate lower than the *x* axis value; for example, for Tool 1: unpatched, ~80% of the templates (of 72 total) have an obscuration rate of <20%. In other words, steeper (more left-shifted) curves indicate a more desirable performance. See also figs. S19 to S23, table S4, and data S6.

to a known POC. Indeed, this hypothesis is supported by the fact that the length of the longest fragment in an obfuscated DNA sequence encoding a synthetic homolog can serve as a decision value in predicting whether DNA obfuscation will lead to an obscuration event (fig. S22).

Conclusions

We believe that the ongoing advancement of AIPD holds great promise for tackling critical challenges in health and the life sciences, with the potential to deliver overwhelmingly positive impacts on people and society (*9*). As with other emerging technologies, however, it is also crucial to proactively identify and mitigate risks arising from novel capabilities. We hope this project serves as a timely example of how responsive and responsible risk mitigation can be achieved through a combination of technical innovation, collaborative exploration, objective analysis, and a deliberative process, thereby contributing to the growing body of work on AI safety (*40*). In the long term, sequence-based biosecurity screening alone is unlikely to remain sufficient for flagging restricted sequences, as we envision a future in which AIPD produces proteins unlike any found in nature. Accordingly, although efforts to improve sequence-based hazard detection software should continue, we must also pursue the development of new approaches. Next steps could include, for example, augmenting existing BSS tools through homology detection using high-dimensional learned embedding spaces (*49–53*).

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Supplementary Text; Data and Materials Availability Plan; Data S1 to S6; Figs. S1 to S25; Tables S1 to S6; References (*54–60*); MDAR Reproducibility Checklist; Appendix S1

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Strengthening nucleic acid biosecurity screening against generative protein design tools

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Editor's summary

One way in which companies and governments can limit potentially harmful biological research is by restricting the commercial synthesis of DNA that encodes particular proteins or is from select pathogens. However, screening methods are not necessarily designed to detect engineered sequences. Wittmann *et al*. worked with four commercial DNA synthesis companies to stress test and develop patches for screening methods to greatly improve their ability to identify sequences that should be restricted. Their results demonstrate that proteins of concern can be flagged by updated software even if their sequences have been altered using protein design methods. —Michael A Funk

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